

## Set Up ROS 2 Repositories

```
sudo apt update
sudo apt install software-properties-common -y
sudo add-apt-repository universe -y

sudo apt install curl -y
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o
/usr/share/keyrings/ros-archive-keyring.gpg

echo "deb [arch=$(dpkg --print-architecture)
signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(.
/etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee
/etc/apt/sources.list.d/ros2.list > /dev/null
```

## Install ROS 2 Jazzy

```
sudo apt update
sudo apt upgrade -y
sudo apt install ros-jazzy-desktop -y
```

## Source the Environment

```
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

## Install the RPLIDAR Package

```
sudo apt update
sudo apt install ros-jazzy-rplidar-ros -y
```

## Grant Serial Port Permissions

```
sudo chmod 666 /dev/ttyUSB0
```

## Execute the RPLIDAR Node

```
ros2 run rplidar_ros rplidar_composition --ros-args -p serial_port:=/dev/ttyUSB0 -p
frame_id:=laser_frame -p angle_compensate:=true -p scan_mode:=Standard
```